

EDUCATION

Bachelors of Science in Professional Physics. Montana State University, 2018. 3.95 GPA

WORK EXPERIENCE

Tonic.ai

Aug 2024 - Present

Senior Frontend Engineer

- Teamed up with a backend engineer to implement role-based access control (RBAC) to a TypeScript React app. This enabled compliance with government security standards and unlocked a contract valued at \$800K
- Improved auditability of PII results by adding in entity preview and correction interfaces into a TypeScript React app. This allowed the team to more effectively market to enterprise customers
- Proactively maintained and patched dozens of frontend NPM dependencies reducing security vulnerabilities and ensuring compliance with enterprise security policies

Emotive.io

Nov 2021 - Jun 2024

Lead Software Engineer

- Worked with finance team to lead a project automating the import + upload + charging of monthly billing spreadsheets. Uploading + parsing + validating a csv using strongly typed Python/Django which prevented the org from doing 100s of hours of manual, error-prone tasks. Uploading the documents to S3 ensured auditability
- Implemented a TCPA litigator list screening feature to allow brands to filter out enrollments by those who would be looking to sue them, effectively saving brands \$1500 per litigator identified
- Updated and implemented a frontend technical screening process which allowed the org to identify cost-effective contractors and ultimately Reduced the team's labor costs by 50-60%
- Guided a team of contractors based overseas. Worked as a liaison between calls with brands and engineers to translate requests into technical requirements. Provided code review + coordinated discussions surrounding best practices

Senior Software Engineer

- Transitioned portions of a Django based monolith to a Flask + FastAPI based microservice architecture to ensure scalability while maintaining a cyclomatic complexity of 5 or lower. This made it possible for teams to function more independently and increased the org's velocity
- Refactored Django systems to reduce query times by 99.9% by introducing covering indices and lateral joins into our PostgreSQL database. This allowed the database to be downsized which lowered operational costs.
- Rewrote legacy React components to remove unnecessary global state + abstractions allowing for a 75% reduction in total lines of code for a microfrontend
- Lead the company's "Frontend Guild" to educate and elevate the organization's frontend engineers. We explored topics like the transition to React 18, when to use various state management solutions, as well as a general best practice consensus.

15Five Inc

Oct 2019 - Nov 2021

Engineering Manager

- Led a team of 4 engineers tasked with maintaining and improving the core 15Five Check-in feature of our app.
- Cultivated a healthy atmosphere by encouraging direct reports to bring their full selves to work through regular 1-on-1's, performance reviews, and role clarity discussions which led to 100% retention of the engineers on my team.
- Collaborated with designers, product managers, mobile engineers, and other teams while participating in customer calls to incorporate feedback, build new features, and integrate other teams' work into the Check-in.

Technical Lead

- Developed end to end solutions for all feature work. We typically began by establishing scope before writing and presenting an implementation plan and tickets. During the development cycle I would write code and review 100% of the relevant pull requests.
- Implemented TypeScript adoption at an org level. I worked with several other developers to enforce static typing on 100% of our React components. This was part of the effort that allowed our DevOps team to embrace CI/CD and decrease cycle time by 50% org-wide.
- Architected and built an extensible framework to simplify app-wide rich text editing. I began by investigating existing usage in order to determine the proper abstractions before removing 1000s of lines of duplicated code and several abstraction layers from our codebase.

Frontend Engineer

- Built OKR tracking feature into existing app infrastructure. This feature has grown into our most adopted feature. In order to implement this, we had to integrate custom React hooks and functional components into legacy jQuery business logic.

Triple Tree LLC

Mar 2019 - Aug 2019

Software Engineer

- Built a veteran transition program with the COMMIT Foundation. Implemented Oauth2 to ensure data security and developed a video streaming and recommendation engine by transforming client designs into React + Redux components.
- Researched, Designed, & Implemented app wide email automation logic, canary releases on AWS, a video recommendation engine built in NodeJS

The Business Garage

Feb 2018 - Mar 2019

Junior Software Engineer

- Built the backend using Django and managed deployments for a web app with 50K users. We experienced viral growth and implemented redis caching to reduce load times by 60%. Prior to the growth, I ensured scalability by developing distributed load test infrastructure through AWS.
- Researched, Designed, & Implemented push notifications through a redis queue, content caching to reduce db stress, distributed load testing on AWS

LIGO Scientific Collaboration

2013, 2016 - 2018

Researcher

- Wrote python applications + received grant funding to develop deep neural networks using Tensorflow which modeled neutron stars to test Einstein's theory of gravity.

OTHER PROJECTS, EXPERIENCE & SKILLS

Skills

- **Frontend:** TypeScript, JavaScript, Webpack, HTML, CSS, Sass, React, Redux, React Hook Form, React Final Form, TanStack (query, table, virtual, etc), jQuery, D3, Tailwind, Bootstrap, some (but limited) angular + vue
- **Backend:** Python, Django, Django REST Framework, Django Channels, Celery, Flask, FastAPI, Node.js, Express.js, AdonisJS, Hapijs, Redis, PostgreSQL, CouchDB (+ some MongoDB), SQLite, MariaDB, RethinkDB
- **DevOps/Cloud:** Docker, Docker Compose, AWS (EKS, Lambda, Gateway, SSM, RDS, Cloudwatch, S3, EBS among others), Kubernetes, Heroku
- **Data Science:** Spacy, TensorFlow, Pandas, Jupyter, SciPy, SKLearn, NLTK, Matplotlib, Seaborn, Bokeh, Keras
- **Other:** Linux, Bash, GNU Make, Git, Mathematica, vim

Open source contributions—  github.com/jpribyl &  johnpribyl.com

- Wrote, published, and maintain react-hook-dragula and react-hook-mathjax which have received around 20 stars and 300 downloads per week. I have also contributed to React, Django, vim-test, and Freqtrade.
- Forked, fixed a deprecation warning, and released action-docker-layer-caching when unable to reach the original maintainer.

Teaching & Coaching

- A combined 8 years of experience coaching soccer and tutoring math and physics up to a college level and have worked with ages ranging from 5 to 50 years and subjects ranging from arithmetic to quantum mechanics.

Grand Canyon Rafting

- Successfully led a 3 week, unguided raft trip down 280 river miles through the canyon. I planned all the logistics for 16 people.